

Dynamics, Trajectory and Planning

Intro to Robotics

1 Torques and Forces on Joints

There are two types of joints: revolute and prismatic.

- Revolute joints produce torque.
- Prismatic joints produce force.

For static force/torque mapping, the standard Jacobian relationship is

$$\tau = J^T(q) W,$$

where

- τ is the vector of generalized joint efforts (torques for revolute joints and forces for prismatic joints),
- W is the end-effector wrench (force and moment vector),
- $J(q)$ is the manipulator Jacobian.

For a planar problem involving only linear forces, W may be replaced by the corresponding force vector F .

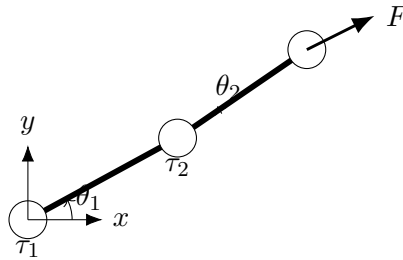


Figure 1: The Jacobian maps joint velocities to end-effector velocity; its transpose maps an end-effector wrench back to generalized joint efforts.

2 Lagrangian Mechanics (Torques and Forces)

The goal is to find the torques or forces acting on a joint.

The Lagrangian is

$$\mathcal{L} = KE - PE,$$

where

$$KE = \frac{1}{2}mv^2$$

and

$$PE = mgh.$$

For the joint variables used in the notes,

- θ represents position,
- $\dot{\theta}$ represents velocity,
- revolute joints correspond to torque,
- prismatic joints correspond to force.

2.1 Algorithm

1. Determine the robot manipulator position vector, represented by the joint coordinates.
2. Differentiate the position vector to obtain the velocity vector.
3. Compute kinetic and potential energy:

$$KE = \frac{1}{2}mv^2, \quad PE = mgh.$$

4. Construct the Lagrangian:

$$\mathcal{L} = KE - PE.$$

5. Use the Euler–Lagrange equation to determine the generalized joint force:

$$Q_i = \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i}.$$

Here Q_i is a torque for a revolute coordinate and a force for a prismatic coordinate. If a known nonconservative external generalized force Q_i^{ext} is separated from the actuator effort u_i , then

$$u_i + Q_i^{\text{ext}} = \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i}.$$

2.2 Worked Example

The example contains two prismatic joints, P_1 and P_2 . The link masses are m_1 and m_2 , the joint displacements are q_1 and q_2 , and the actuator input forces are F_1 and F_2 .

2.2.1 Position Vectors

The mass position vectors shown in the notes are

$$P_{m_1} = \begin{bmatrix} 0 \\ q_1 \end{bmatrix}, \quad P_{m_2} = \begin{bmatrix} q_2 \\ q_1 \end{bmatrix}.$$

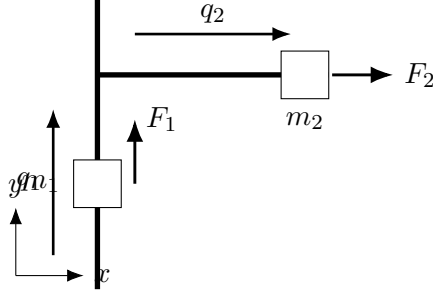


Figure 2: Two-prismatic-joint mechanism used in the Lagrangian worked example. The vertical coordinate is q_1 and the horizontal coordinate of m_2 is q_2 .

2.2.2 Velocity Vectors

Velocity is obtained by differentiating the position vector:

$$V_m = \frac{dP_m}{dt}.$$

Therefore,

$$V_{m_1} = \begin{bmatrix} 0 \\ \dot{q}_1 \end{bmatrix}, \quad V_{m_2} = \begin{bmatrix} \dot{q}_2 \\ \dot{q}_1 \end{bmatrix}.$$

2.2.3 Kinetic Energy

Starting with

$$KE = \frac{1}{2}mv^2,$$

and, for a planar velocity vector,

$$KE = \frac{1}{2}m(v_x^2 + v_y^2),$$

the kinetic energies are

$$KE_{m_1} = \frac{1}{2}m_1\dot{q}_1^2$$

and

$$KE_{m_2} = \frac{1}{2}m_2(\dot{q}_2^2 + \dot{q}_1^2).$$

2.2.4 Potential Energy

Using

$$PE = mgh,$$

where the height is the vertical component, the notes give

$$PE_{m_1} = m_1gq_1$$

and

$$PE_{m_2} = m_2gq_1.$$

2.2.5 The Lagrangian

The Lagrangian is

$$\mathcal{L} = KE - PE.$$

For the two masses,

$$\mathcal{L} = KE_{m_1} + KE_{m_2} - PE_{m_1} - PE_{m_2}.$$

Substituting the expressions above gives

$$\mathcal{L} = \frac{1}{2}m_1\dot{q}_1^2 + \frac{1}{2}m_2(\dot{q}_2^2 + \dot{q}_1^2) - m_1gq_1 - m_2gq_1.$$

This can also be grouped as

$$\mathcal{L} = m_1\left(\frac{1}{2}\dot{q}_1^2 - gq_1\right) + m_2\left(\frac{1}{2}\dot{q}_2^2 + \frac{1}{2}\dot{q}_1^2 - gq_1\right).$$

2.2.6 Lagrange Equations for the Joint Forces

For a generalized coordinate q_i , the Euler–Lagrange equation is

$$Q_i = \frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{q}_i}\right) - \frac{\partial\mathcal{L}}{\partial q_i}.$$

Because q_1 and q_2 are prismatic coordinates, Q_1 and Q_2 have units of force.

For q_1 ,

$$\frac{\partial\mathcal{L}}{\partial\dot{q}_1} = (m_1 + m_2)\dot{q}_1,$$

so

$$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{q}_1}\right) = (m_1 + m_2)\ddot{q}_1.$$

Also,

$$\frac{\partial\mathcal{L}}{\partial q_1} = -(m_1 + m_2)g.$$

Therefore,

$$Q_1 = (m_1 + m_2)\ddot{q}_1 + (m_1 + m_2)g.$$

For q_2 ,

$$\frac{\partial\mathcal{L}}{\partial\dot{q}_2} = m_2\dot{q}_2, \quad \frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{q}_2}\right) = m_2\ddot{q}_2,$$

and

$$\frac{\partial\mathcal{L}}{\partial q_2} = 0.$$

Thus,

$$Q_2 = m_2\ddot{q}_2.$$

If F_1 and F_2 denote the actuator forces and there are no other nonconservative generalized forces, then

$$F_1 = Q_1, \quad F_2 = Q_2.$$

Hence,

$$F_1 = (m_1 + m_2)(\ddot{q}_1 + g)$$

and

$$\boxed{F_2 = m_2 \ddot{q}_2}.$$

The terms $F_1 q_1$ and $F_2 q_2$ that appeared in the handwritten derivation are not subtracted directly in the Euler–Lagrange equation when F_1 and F_2 are actuator forces. A product such as Fq has units of work/energy, not force. External nonconservative forces instead enter as generalized forces on the right-hand side of the Euler–Lagrange equation.

3 Trajectory Planning and Generation

A trajectory is the time history of position, velocity, and acceleration for each degree of freedom (DOF).

Important considerations include

- run time,
- path update rate (approximately 60–2000 Hz),
- smooth motion in order to minimize wear and tear,
- via points, which are auxiliary/intermediate points between the initial and final positions.

Trajectory planning can be viewed as an optimization problem because there are many possible ways to move from an initial position to a final position.

3.1 Path Generation Methods

3.1.1 Joint-Space Schemes

Joint-space schemes use functions of the joint angles. Each via point, including the initial and final points, is converted to a set of joint angles using inverse kinematics.

3.1.2 Cartesian Schemes

Cartesian schemes directly use the position and orientation of the robot frames together with path points specified by the user to generate the trajectory.

In general, the notes state that joint-space schemes are easier to use and compute because Cartesian schemes require inverse kinematics at every point along the path during the path update, increasing the computational burden on the robot.

3.2 Joint-Space Schemes

The notes divide joint-space schemes into two methods:

- cubic polynomials (or even higher-order polynomials),
- parabolic blends.

3.3 Cubic Polynomials

The idea is to find a third-degree polynomial satisfying the trajectory constraints.

A standard point-to-point cubic trajectory uses four boundary constraints:

$$\theta(t_0) = \theta_0, \quad \dot{\theta}(t_0) = \dot{\theta}_0,$$

$$\theta(t_f) = \theta_f, \quad \dot{\theta}(t_f) = \dot{\theta}_f.$$

For the common case in which the joint starts and stops at rest,

$$\dot{\theta}_0 = \dot{\theta}_f = 0.$$

A cubic polynomial has four coefficients, so four independent constraints are sufficient to determine them. If via points are added, the degree of the polynomial can be increased by adding additional constraints.

The position function is

$$\theta(t) = a_0 + a_1t + a_2t^2 + a_3t^3.$$

The velocity is

$$\dot{\theta}(t) = a_1 + 2a_2t + 3a_3t^2.$$

The acceleration is

$$\ddot{\theta}(t) = 2a_2 + 6a_3t.$$

3.3.1 Cubic Polynomial Worked Example

The image shows a motion through three points

$$A \rightarrow B \rightarrow C.$$

For point A ,

$$\theta_0 = 2^\circ, \quad \dot{\theta}_0 = 0.$$

The via point is shown at approximately

$$\theta_v = 15^\circ.$$

For point C ,

$$\theta_f = 20^\circ, \quad \dot{\theta}_f = 0.$$

The diagram shows a time interval of 2 for each segment.

Segment $A \rightarrow B$. Using

$$\theta(t) = a_0 + a_1t + a_2t^2 + a_3t^3,$$

the initial position condition gives

$$\theta(0) = a_0 + a_1(0) + a_2(0)^2 + a_3(0)^3 = 2,$$

so

$$a_0 = 2.$$

The initial velocity condition gives

$$\dot{\theta}(0) = a_1 + 2a_2(0) + 3a_3(0)^2 = 0,$$

so

$$a_1 = 0.$$

At $t = 2$,

$$\theta(2) = 2 + a_2(2)^2 + a_3(2)^3 = 15,$$

which gives

$$4a_2 + 8a_3 = 13.$$

The velocity condition shown in the image gives

$$\dot{\theta}(2) = 2a_2(2) + 3a_3(2)^2,$$

leading to

$$4a_2 + 12a_3 = 0.$$

Solving the equations gives the coefficients shown in the notes:

$$a_0 = 2, \quad a_1 = 0, \quad a_2 = 9.75, \quad a_3 = -3.25.$$

Therefore,

$$\boxed{\theta(t) = 2 + 9.75t^2 - 3.25t^3} \quad (A \rightarrow B).$$

Segment $B \rightarrow C$. The coefficients shown for the second segment are

$$b_0 = 15, \quad b_1 = 0, \quad b_2 = 3.75, \quad b_3 = -1.25.$$

Therefore,

$$\boxed{\theta(t) = 15 + 3.75t^2 - 1.25t^3} \quad (B \rightarrow C).$$

3.4 Parabolic Blends

The notes state that parabolic blends are used to solve the problem of discontinuous velocities.

The blend uses constant acceleration, and the two blend times are shown as equal:

$$t_b = t_{\text{blend 1}} = t_{\text{blend 2}}.$$

For the first blend,

$$\boxed{\theta_b = \theta_0 + \frac{1}{2}\ddot{\theta} t_b^2}.$$

The blend time is shown as

$$\boxed{t_b = \frac{t_f}{2} - \frac{\sqrt{\ddot{\theta}^2 t_f^2 - 4\ddot{\theta}(\theta_f - \theta_0)}}{2\ddot{\theta}}}.$$

The acceleration constraint shown in the image is

$$\boxed{|\ddot{\theta}| \geq \frac{4|\theta_f - \theta_0|}{t_f^2}}.$$

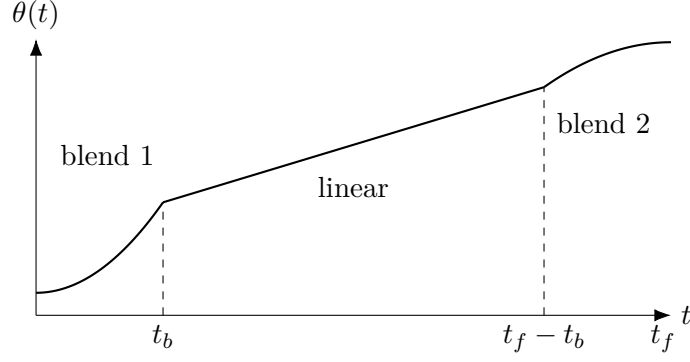


Figure 3: Conceptual parabolic-blend trajectory: acceleration is used near the beginning and end, with a constant-velocity linear segment between them.

The trajectory is divided into three regions: the first parabolic blend, a linear section, and the second parabolic blend.

The first blend is

$$\theta(t) = \theta_0 + \frac{1}{2}\ddot{\theta}t^2, \quad 0 < t < t_b.$$

The velocity at the end of the first blend is

$$\dot{\theta}_b = \ddot{\theta}t_b,$$

and the position there is

$$\theta_b = \theta_0 + \frac{1}{2}\ddot{\theta}t_b^2.$$

To maintain position continuity, the linear section is

$$\theta(t) = \theta_b + \dot{\theta}_b(t - t_b), \quad t_b < t < t_f - t_b.$$

The second blend is

$$\theta(t) = \theta_f - \frac{1}{2}\ddot{\theta}(t - t_f)^2, \quad t_f - t_b < t < t_f.$$

Thus, the piecewise trajectory written in the image is

$$\theta(t) = \begin{cases} \theta_0 + \frac{1}{2}\ddot{\theta}t^2, & 0 < t < t_b, \\ \theta_b + \dot{\theta}_b(t - t_b), & t_b < t < t_f - t_b, \\ \theta_f - \frac{1}{2}\ddot{\theta}(t - t_f)^2, & t_f - t_b < t < t_f. \end{cases}$$

4 Review Sheet

Robot Dynamics and Trajectory Planning Lecture Notes

1 Robot Manipulator Dynamics

1.1 Manipulator Model

A robot manipulator is a multi-body system with joints and links. The dynamics relate the joint torques/forces τ to the motion $q, \dot{q}, \ddot{q}$.

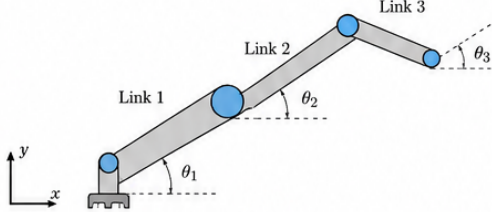


Figure 1: Serial manipulator with three revolute joints.

The equations of motion can be written in the standard form

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = \tau + \tau_{\text{ext}} \quad (1)$$

where

- $M(q) \in \mathbb{R}^{n \times n}$: inertia (mass) matrix (symmetric, positive definite)
- $C(q, \dot{q})\dot{q}$: Coriolis and centrifugal terms
- $g(q)$: gravitational forces/torques
- τ : vector of joint torques/forces (control input)
- τ_{ext} : external torques/forces (e.g., from contact)

2 Lagrangian Formulation

2.1 Euler-Lagrange Equation

Define the Lagrangian as the difference between total kinetic and potential energy:

$$\mathcal{L}(q, \dot{q}) = T(q, \dot{q}) - V(q) \quad (2)$$

The equations of motion are obtained from the Euler-Lagrange equation:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q_i \quad (3)$$

where Q_i is the generalized force (torque for a revolute joint, force for a prismatic joint).

2.2 Kinetic and Potential Energy

Kinetic Energy

$$T = \frac{1}{2} m v^2 \quad (4)$$

For planar motion:

$$T = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) \quad (5)$$

For a rigid body:

$$T = \frac{1}{2} m v_{\text{cm}}^2 + \frac{1}{2} \omega^T I \omega \quad (6)$$

Potential Energy

For gravity (height h):

$$V = mgh \quad (7)$$

For a spring (stiffness k):

$$V = \frac{1}{2} k x^2 \quad (8)$$

3 Worked Example: Two-Prismatic-Joint Manipulator

3.1 System Description

Consider a planar manipulator with two prismatic joints and point masses m_1 and m_2 .

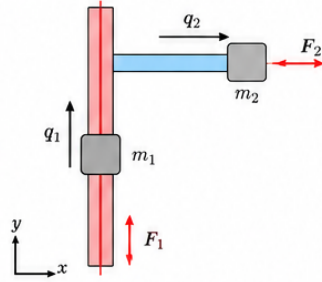


Figure 2: Two-prismatic-joint manipulator.

3.2 Position and Velocity

Position vectors:

$$p_{m1} = \begin{bmatrix} 0 \\ q_1 \end{bmatrix}, \quad p_{m2} = \begin{bmatrix} q_2 \\ 0 \end{bmatrix}$$

Velocity vectors:

$$v_{m1} = \begin{bmatrix} 0 \\ \dot{q}_1 \end{bmatrix}, \quad v_{m2} = \begin{bmatrix} \dot{q}_2 \\ 0 \end{bmatrix}$$

3.3 Energies

Kinetic energy:

$$T = \frac{1}{2} m_1 \dot{q}_1^2 + \frac{1}{2} m_2 (\dot{q}_1^2 + \dot{q}_2^2) \quad (9)$$

Potential energy (gravity):

$$V = (m_1 + m_2) g q_1 \quad (10)$$

Lagrangian:

$$\mathcal{L} = T - V = \frac{1}{2} (m_1 + m_2) \dot{q}_1^2 + \frac{1}{2} m_2 \dot{q}_2^2 - (m_1 + m_2) g q_1 \quad (11)$$

3.4 Equations of Motion

Using the Euler-Lagrange equation:

For q_1 :

$$(m_1 + m_2) \ddot{q}_1 + (m_1 + m_2) g = F_1 \quad (12)$$

For q_2 :

$$m_2 \ddot{q}_2 = F_2 \quad (13)$$

These are the required actuator forces to achieve the motion (assuming no other external forces).

Remark

In general, external forces can be included on the right-hand side as generalized forces Q_i . The terms $F_1 q_1$ and $F_2 q_2$ should not be subtracted as in the original notes; instead, actuator forces appear directly as the generalized forces.

4 Jacobian and Force Relationships

4.1 End-Effector Velocity

The end-effector linear velocity v is related to joint velocities by

$$v = J(q)\dot{q} \quad (14)$$

4.2 Force Transformation

The relationship between end-effector wrench W (forces/moments) and joint torques/forces τ is

$$\tau = J(q)^T W \quad (15)$$

where $W = [f^T \ n^T]^T$ is the wrench in Cartesian space.

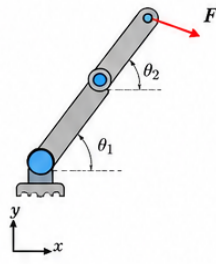


Figure 3: End-effector force and joint torques.

5 Trajectory Planning

5.1 Cubic Polynomial Trajectory

A cubic polynomial has the form

$$\theta(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 \quad (16)$$

with

$$\dot{\theta}(t) = a_1 + 2a_2 t + 3a_3 t^2 \quad (17)$$

$$\ddot{\theta}(t) = 2a_2 + 6a_3 t \quad (18)$$

Boundary conditions (four constraints):

$$\begin{cases} \theta(0) = \theta_0 \\ \dot{\theta}(0) = \dot{\theta}_0 \\ \theta(t_f) = \theta_f \\ \dot{\theta}(t_f) = \dot{\theta}_f \end{cases} \quad (19)$$

Solving gives unique coefficients a_0, a_1, a_2, a_3 .

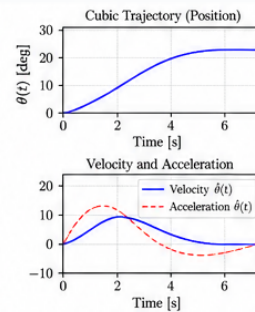


Figure 4: Example cubic trajectory with zero initial and final velocity.

5 Credit

These notes are based on the robotics playlist referenced in the original notes as the YouTube channel `@thatsengineering5235`. The original notes state that the external images are examples from those videos.